

Software Engineering



National Transmission & Despatch Company

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Introduction

The National Transmission & Despatch Company (NTDC) is responsible for transmitting electricity across Pakistan.

It faces a major challenge of high transmission losses, which lead to inefficiencies and frequent power outages.

This aims to analyze the causes of these losses and propose solutions to improve NTDC's efficiency and reliability in power transmission.



Problem statement and Root cause

NTDC faces high transmission losses due to

- outdated infrastructure
- inadequate maintenance
- Outdated software / lack of real time updates

Results in

- power outages
- increased costs
- impacting the reliability of the electricity supply.

StakeHolder identification

Internal Stakeholder

01

Employees

02

Management

03

Board of Directors

External Stakeholder

01

Government Agencies

02

Local Communities

03

Regulatory Bodies

04

Investors and Financial
Institutions

Solution System Boundaries and Constraints

Solution System Boundaries

- Real-Time Monitoring
- Infrastructure upgrades
- Advanced Data Analytics
- Maintenance Practices

Constraints

- Budget Limitations
- Organizational Policies
- Technological Restrictions

User Needs

- Reliable power transmission
- Reduced outages
- improved monitoring capabilities for timely issue resolution.

User Needs → Features

- Real-Time Monitoring
- Predictive Maintenance
- Advanced Data Analytics
- Automated Reporting

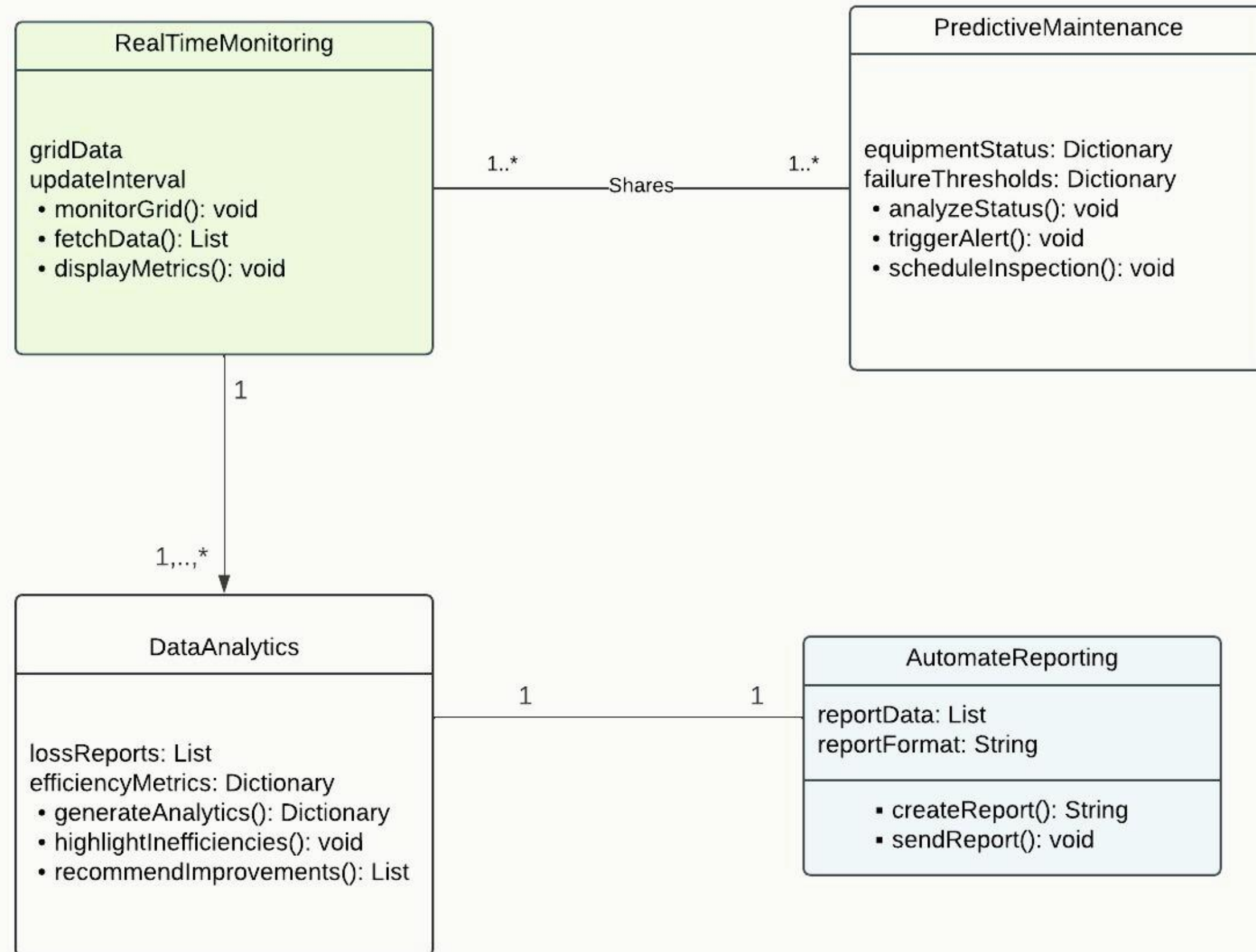
Functional Requirements

Functional Requirement	Description
Real-Time Monitoring	Tracks grid performance live.
Predictive Maintenance	Alerts for potential equipment failures.
Data Analytics	Identifies inefficiencies and high-loss areas.
Automate Reporting	Generates detailed transmission reports.

Non Functional Requirements

Non Functional Requirement	Description
Reliability	99.9% uptime
Scalability	Handles expansion in grid coverage
User-Friendliness	Intuitive UI for NTDC operators.
Data Security	Encryption and access control for data.

Class Diagram



Define Project Scope

01 Real-Time Monitoring:

Continuously tracks network performance, providing instant updates on transmission lines and equipment status.

02 Predictive Maintenance

Uses data analytics to predict equipment failures, allowing for proactive maintenance and reducing unplanned outages.

03 Data Analytics

Analyses transmission data to identify inefficiencies and high-loss areas, enabling targeted improvements.

04 Automated Reporting

Generates regular reports on performance, outages, and maintenance for stakeholders, ensuring transparency and efficient communication.

Define Project Out of Scope

01 Distribution Company Inefficiencies

This solution will not tackle the operational problems or financial issues of distribution companies that contribute to circular debt.

03 Environmental Regulations

will not focus on following environmental laws or dealing with land acquisition issues, although these may need separate attention

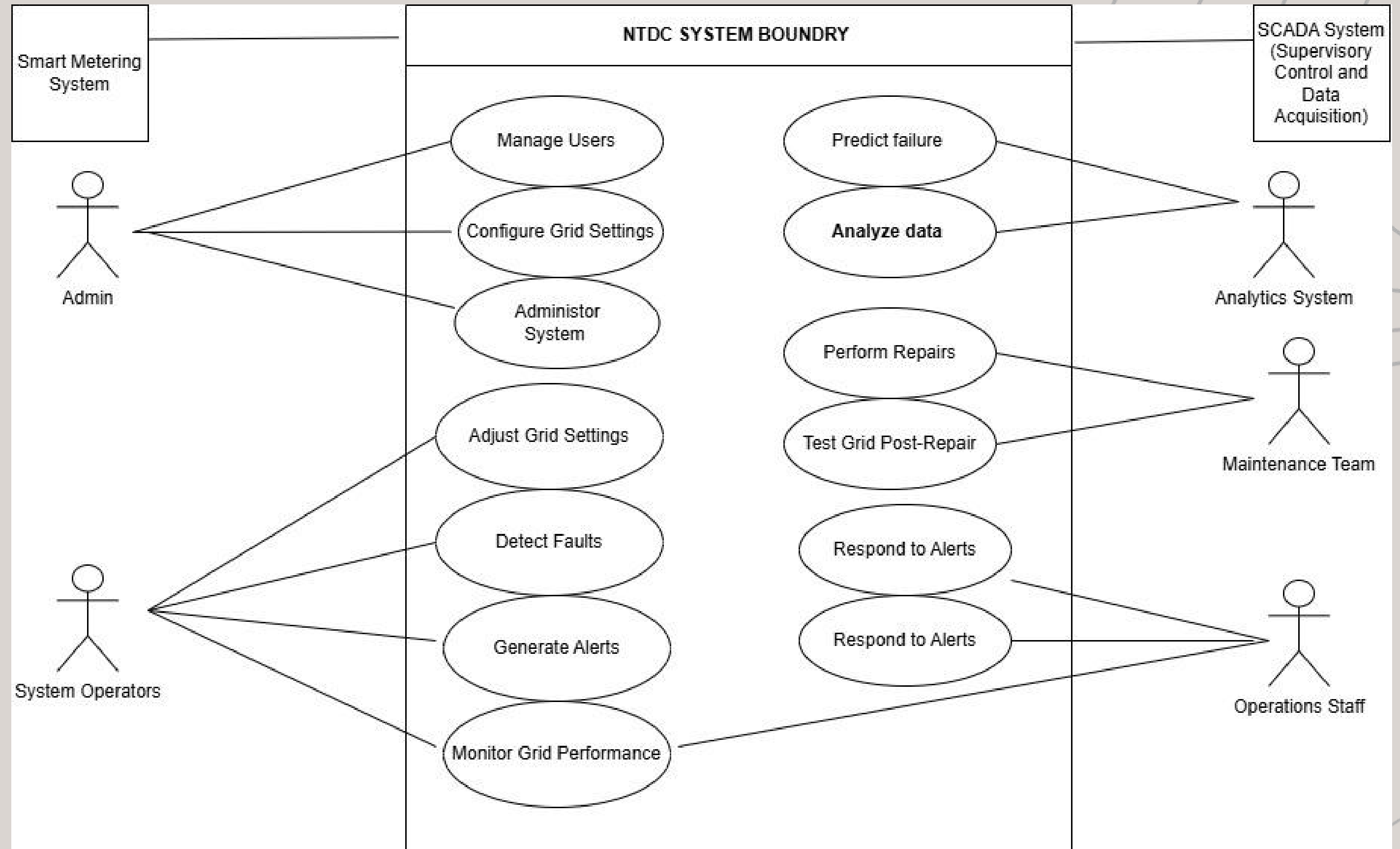
02 Consumer Issues

Problems related to energy theft or customer dissatisfaction at the distribution level will not be addressed.

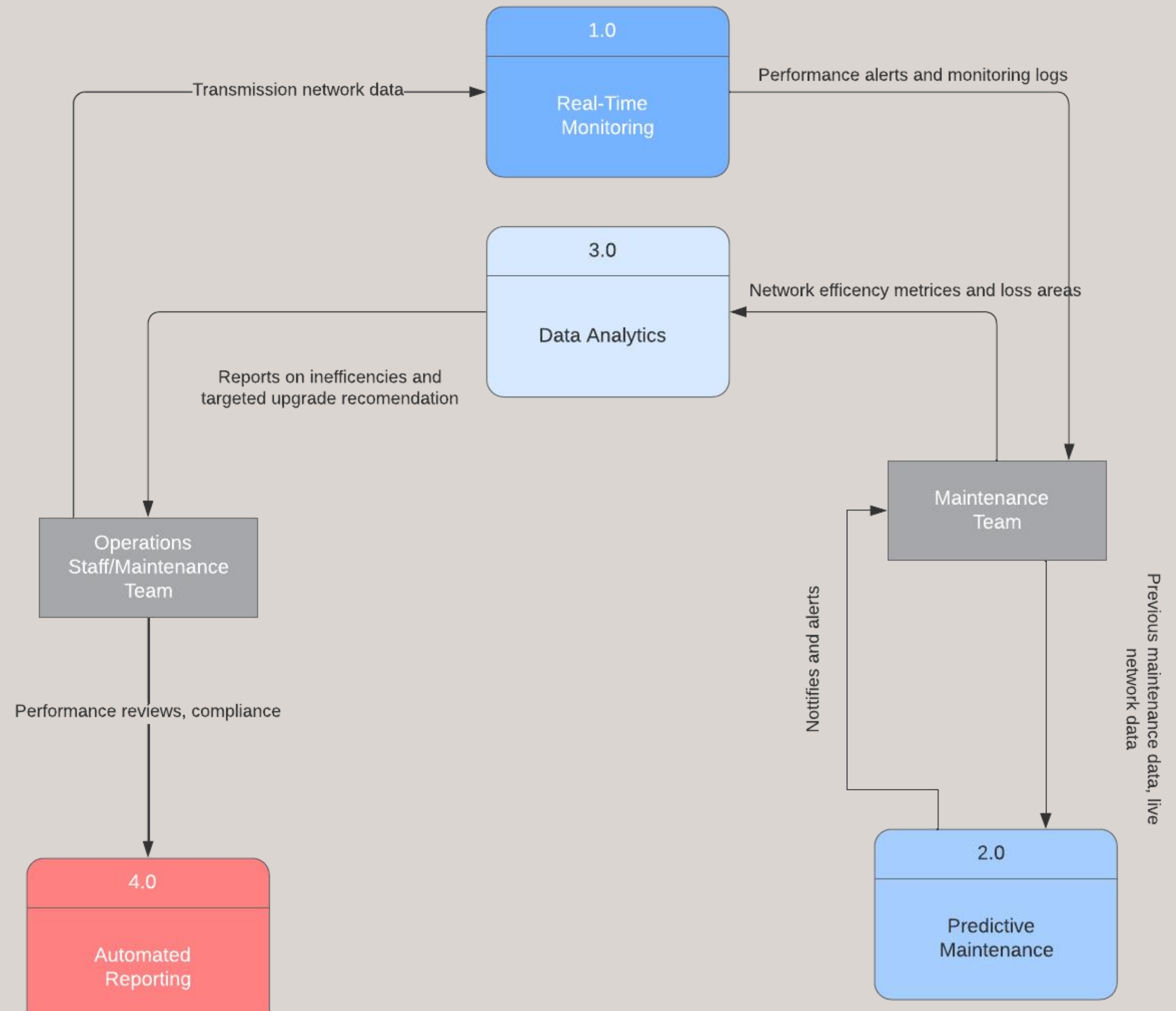
04 Political Influences

will not address political factors that may impact decision making and project execution.

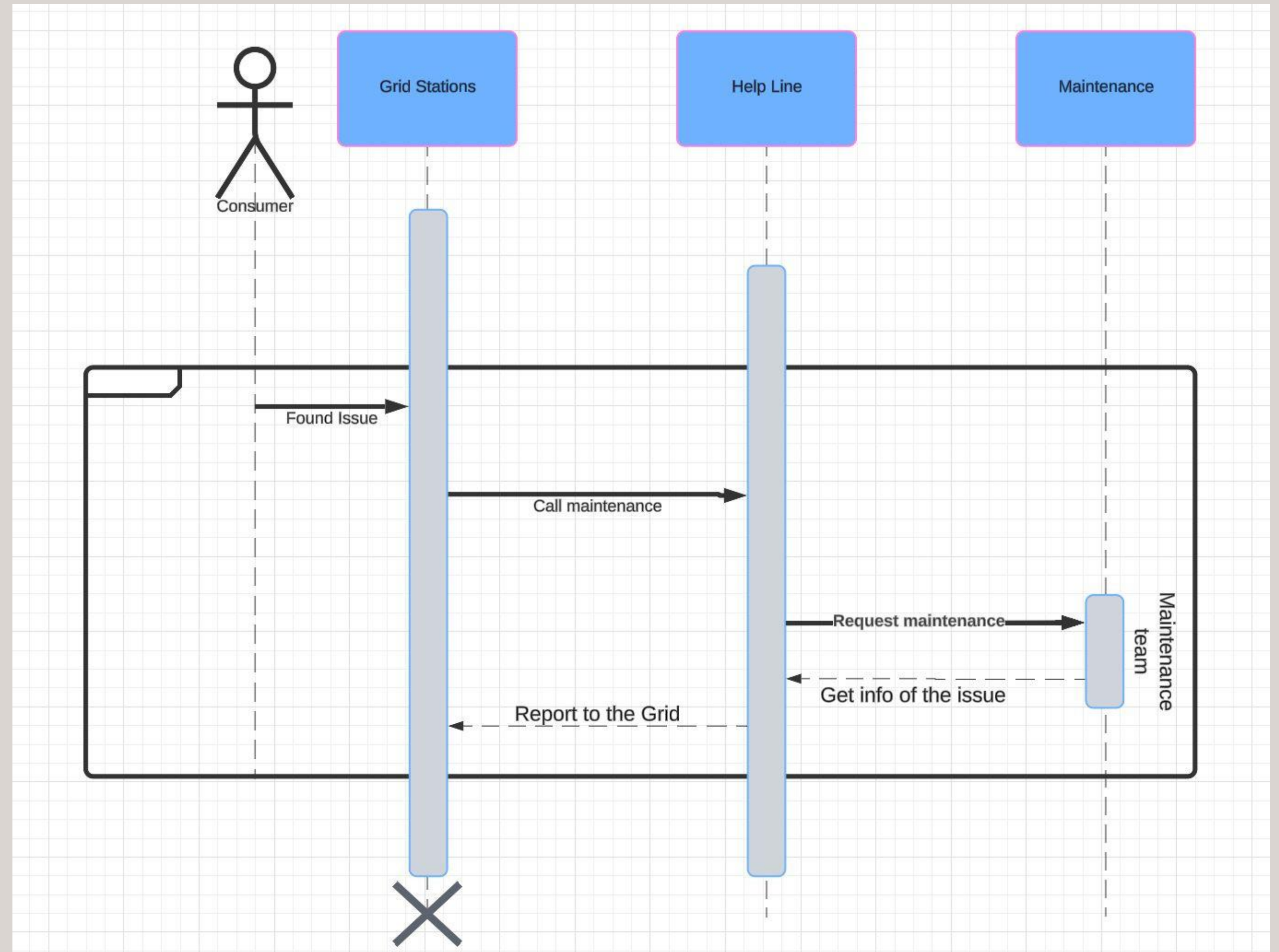
Use Cases



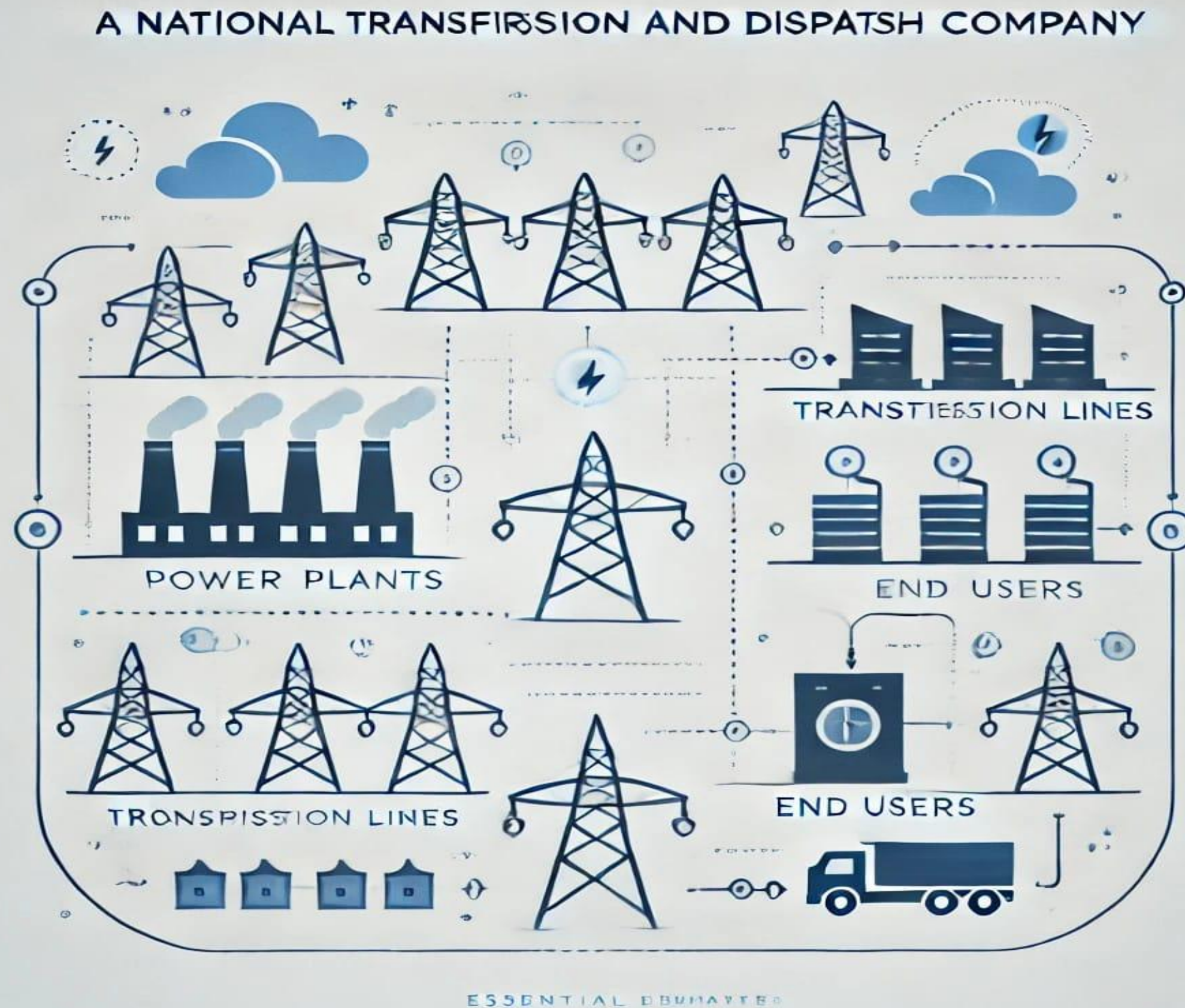
Data Flow Diagram



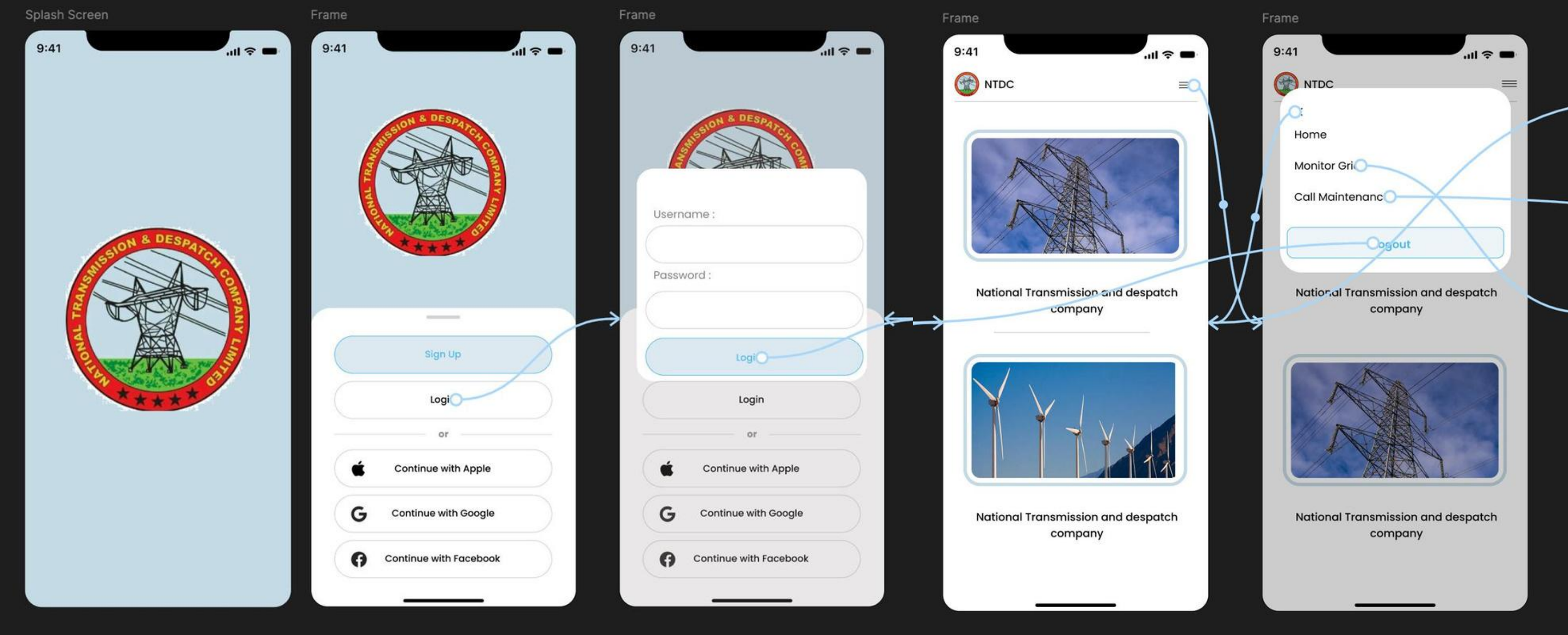
Sequence Diagram



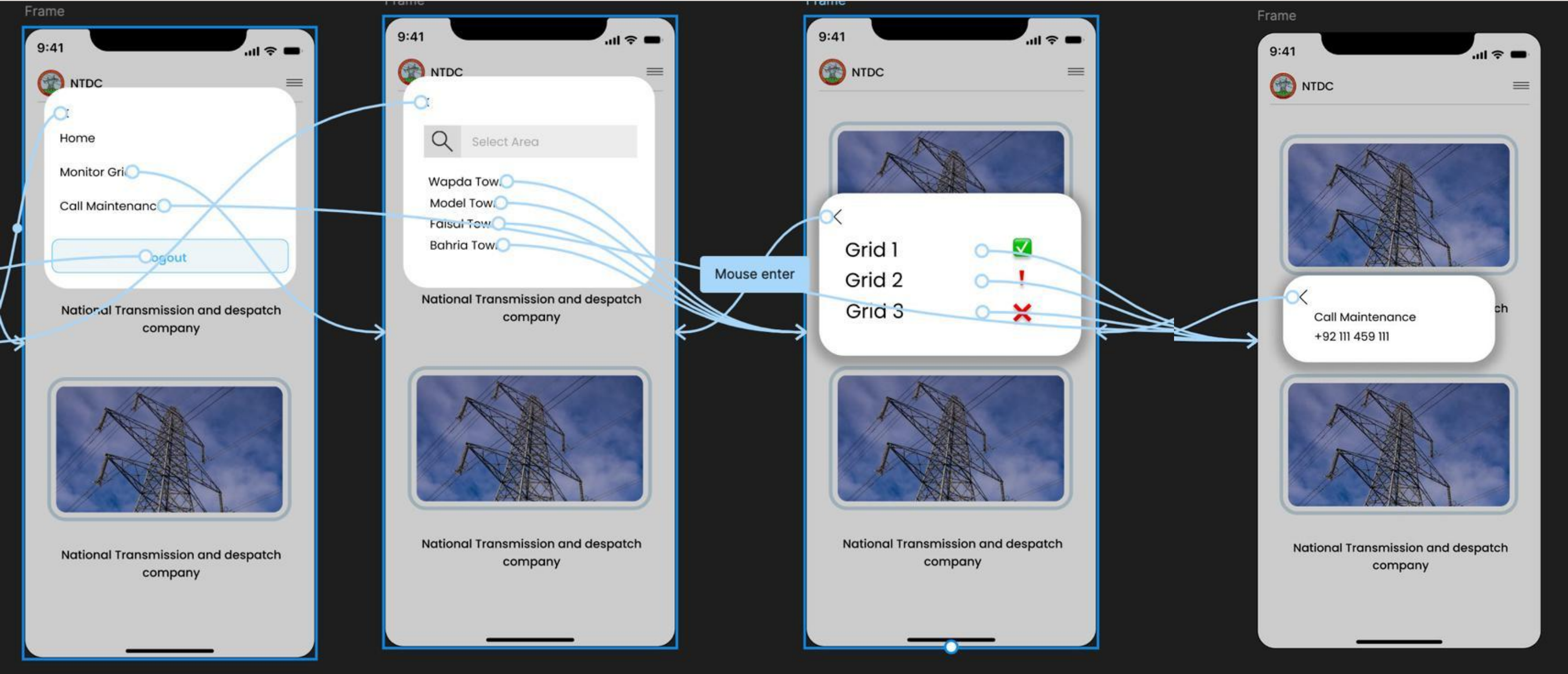
Contribution Diagram

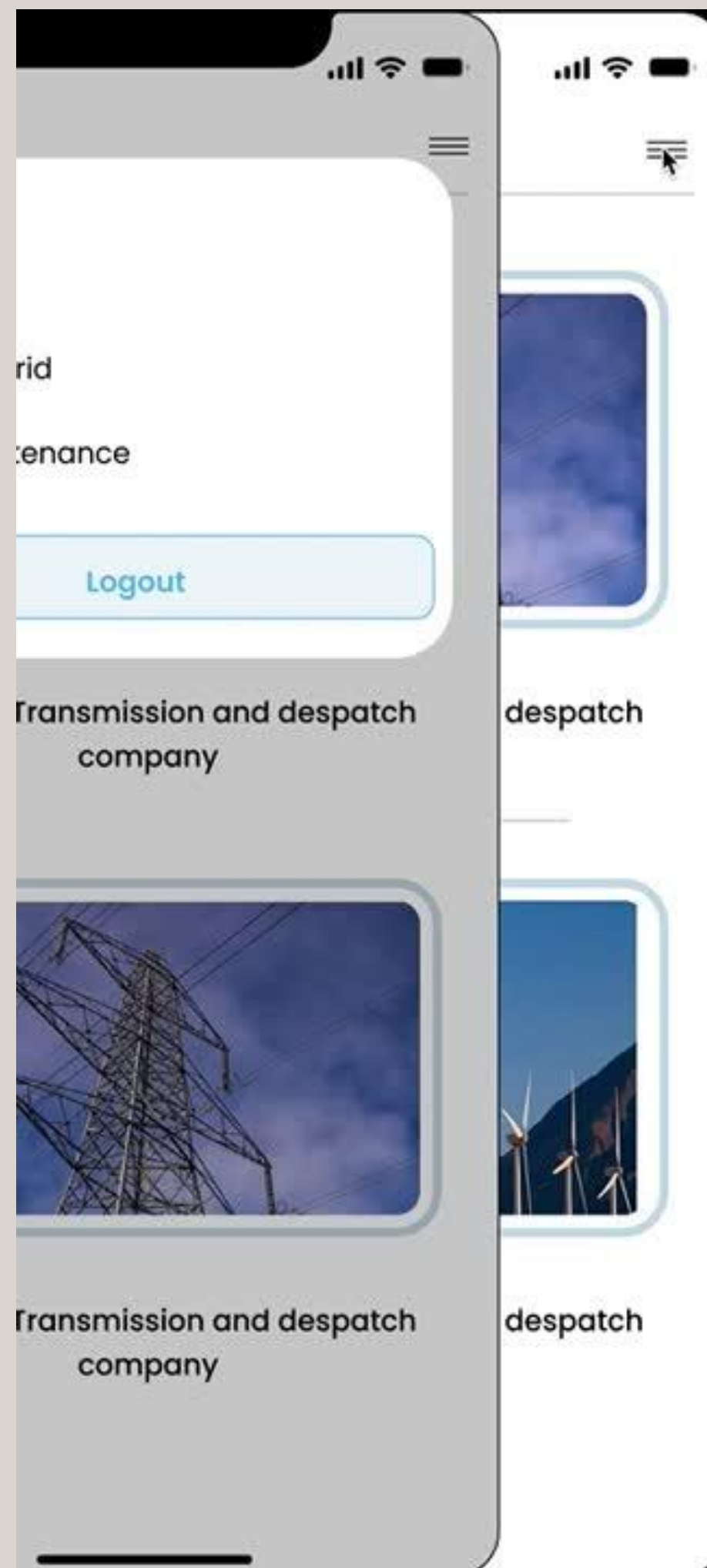


Graphic User Interface



Graphic User Interface





Conclusion

The proposed system for NTDC addresses key challenges of transmission losses and outages by focusing on real-time monitoring, predictive maintenance, and data analytics. The project scope aligns with the **team's skills**, resources, and timeline, ensuring feasibility. Any over-scope issues can be managed by scaling back non-critical features.

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**Thank
You**